1. Attendance Recovery Computation (Question 12)

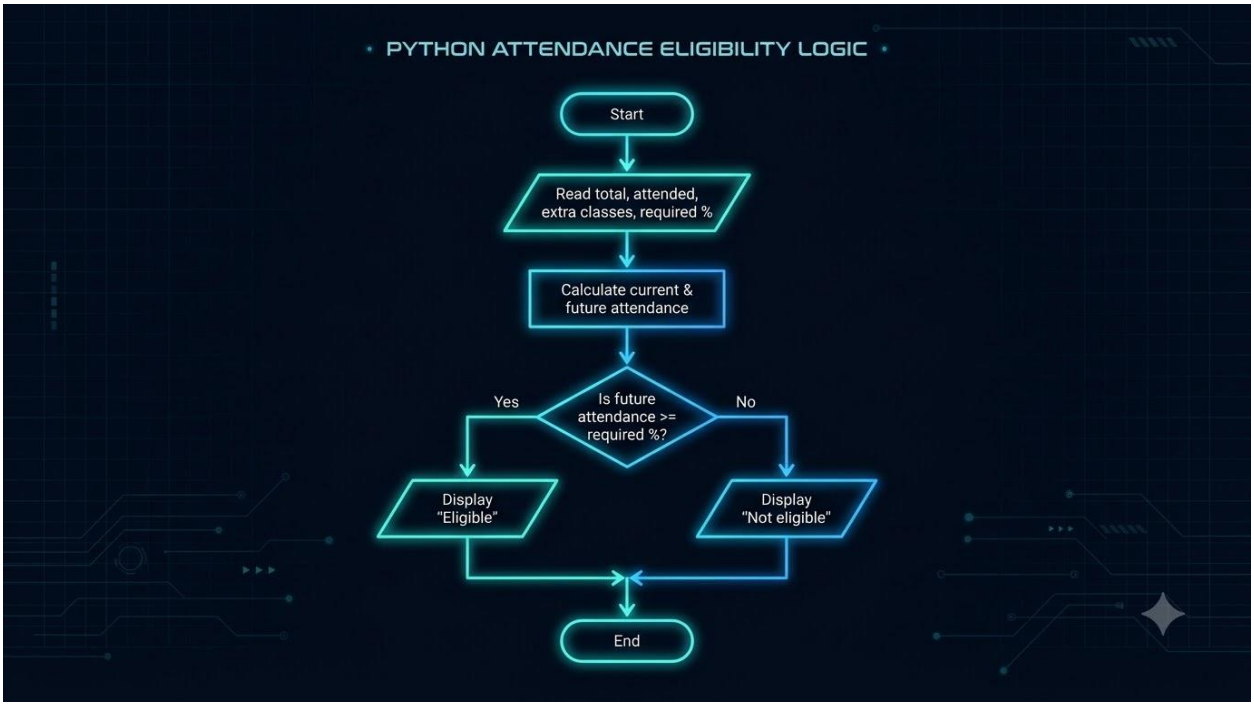
Computational thinking

- Input: Total classes, attended classes and planned additional classes.
- Process: Calculate current attendance and the attendance after attending additional classes.
- Output: Current attendance, future attendance and whether the required attendance is achieved.
- Decomposition: Separate the calculation into current attendance and future attendance.

Algorithm

1. Read total classes, attended classes and additional classes.
2. Read the required attendance percentage.
3. Calculate current attendance.
4. Calculate future attendance after attending all additional classes.
5. Compare future attendance with the required percentage.
6. Display the result.

Flowchart



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Pseudocode

START

INPUT total_classes, attended_classes

INPUT extra_classes, required_percentage

current = attended_classes / total_classes * 100

future = (attended_classes + extra_classes) /

(total_classes + extra_classes) * 100

DISPLAY current, future

IF future >= required_percentage

DISPLAY "Required attendance achieved"

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ELSE
   DISPLAY "Required attendance not achieved"
END IF
STOP
```

Python program

```
total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))
extra_classes = int(input("Enter additional classes to attend: "))
required_percentage = float(input("Enter required attendance percentage: "))

if total_classes <= 0 or attended_classes < 0 or extra_classes < 0:
   print("Invalid input")
elif attended_classes > total_classes:
   print("Attended classes cannot exceed total classes")
else:
   current_attendance = (attended_classes / total_classes) * 100

   future_attended = attended_classes + extra_classes
   future_total = total_classes + extra_classes

   future_attendance = (future_attended / future_total) * 100

   print("Current attendance:", round(current_attendance, 2), "%")
   print("Future attendance:", round(future_attendance, 2), "%")

   if future_attendance >= required_percentage:
```

```
        print("Required attendance can be achieved")
    else:
        print("Required attendance cannot be achieved")
```

Sample output

```
Enter total classes conducted: 100
Enter classes attended: 70
Enter additional classes to attend: 20
Enter required attendance percentage: 75
Current attendance: 70.0 %
Future attendance: 75.0 %
Required attendance can be achieved
```